

Name

Roll No

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI

(An ISO 9001:2015 Certified College)

Affiliated to APJ Abdul Kalam Technological University



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

First Internal Examination – November 2022

FIFTH SEMESTER – B. TECH (A&B)

CST 307: Microprocessors and Microcontrollers

Time: 2 Hours.

Max.Marks:50

PART - A (Answer all Questions)

Qn. No.	Questions	Marks	K level of Qn.	CO(K level)
1	Draw the structure of 8086 flag register and mention the purpose of each flag.	3	K2	CO1(K2)
2	Explain <i>any</i> 3 maximum mode pins of 8086 microprocessor and its functions.	3	K2	CO1(K2)
3	Assume that current value in register AX is FFFDH and carry flag value is 0. What will be the new value in AX register after the execution of each of the following 8086 instructions: a) RCR AX, 01H b) ROR AX, 01H c) SHR AX, 01H	3	K3	CO2(K3)
4	Identify the addressing mode and calculate the effective address for the following instructions with the given values: DS= 1000H, SI= 3000H, BX= 2000H, AX=1500H a) MOV AX,[2040H] b) MOV AX, 50H [BX][SI] c) INC [2000H]	3	K3	CO2(K3)
5	Explain the stack structure of 8086 microprocessor.	3	K2	CO3(K3)

PART - B (Answer any 2 Questions)

Qn No.	Questions	Marks	K level of Qn.	CO & its K level
6	Draw and explain the internal architecture of 8085 microprocessor.	7	K2	CO1(K2)
7	BIU along with the EU forms a pipeline. Justify.	7	K2	CO1(K2)
8	Describe the physical memory organization of 8086.	7	K2	CO1(K2)

PART C (Answer any 2 Questions)

✓	Explain the machine language instruction formats used in 8086.	7	K2	CO2(K3)
✓	Discuss addressing modes supported by 8086 with suitable examples.	7	K2	CO2(K3)
11	List out and explain the string manipulation instructions and unconditional branch instructions with examples. <i>REP, MOVS, MOVSW, CMPS, SCAS, LODS, STOS</i> <i>CALL, RET, INT N, INTO, JMP, IRET, LOOP</i>	7	K2	CO2(K3)

PART D (Mandatory)

12	a) Discuss about the steps followed by 8086 when it detects an interrupt signal in the INTR pin.	4	K2	CO3(K3)
	b) The starting address for a Type 4 interrupt is 1112:1314. Where and in what order this address should be placed in Interrupt Vector Table (IVT).	3	K3	CO3(K3)

14	0000:0000
13	11
12	12
11	13

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Time: 2 Hours.

Max.Marks:50

PART - A (Answer all Questions)

Qn. No.	Questions	Marks	K level (KL)	CO & KL
1 ✓	Identify the functions of the following pins in 8259(PIC): i) $\overline{CAS_0} - CAS_2$ ii) $\overline{SP/EN}$	3	K1	CO3(K3)
2 ✓	Interpret the mode and configurations of 8255 after its control word register is loaded with 84H.	3	K2	CO4(K2)
3 ✓	Sketch the architecture of 8253 Timer.	3	K2	CO4(K2)
4	List out the types of Microcontrollers.	3	K1	CO5(K2)
5 ✓	Explain the memory organization of 8051.	3	K2	CO5(K2)

PART - B (Answer any 2 Questions)

Qn No.	Questions	Marks	K level(KL)	CO&KL
6	Describe the interrupt sequence of 8086 with 8259.	7	K2	CO3(K3)
7 ✓	Explain internal architecture of 8259.	7	K2	CO3(K3)
8 ✓	State the purpose of IVT of 8086 and explain its structure.	7	K2	CO3(K3)

PART C (Answer any 2 Questions)

9 ✓	Detail out the architecture of Programmable Input-Output chip.	7	K2	CO4(K2)
10	a) Explain control word register (CWR) of 8253.	3	K2	CO4(K2)
	b) Discuss the registers used in DMA controller.	4		
11	Draw and explain the architecture of 8257.	7	K2	CO4(K2)

PART D (Mandatory)

12 ✓	Explain the internal architecture of 8051.	7	K2	CO5(K2)
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89 - Int
85 - D
83 - Timer
8051 - MC

Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: CST 307**Course Name: MICROPROCESSORS AND MICROCONTROLLERS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|------|--|---|
| 1 ✓ | What is pipelined architecture? How is it implemented in 8086? | 3 |
| 2 ✓ | Compare the architectural and signal difference between 8086 and 8088. | 3 |
| 3 ✓ | Write any three addressing mode of 8086 with example and write the effective address calculation in each. | 3 |
| 4 ✓ | Write the functions performed by PUSH and POP instructions in 8086 with appropriate diagram. | 3 |
| 5 ✓ | What is an interrupt vector table? Explain its structure in 8086. | 3 |
| 6 ✓ | Write notes on the following based on 8086:
a. software interrupt
b. hardware interrupt
c. nested interrupt | 3 |
| 7 ✓ | Write the function of the following control signals in 8255.
RD, WR, <u>A₀</u> , <u>A₁</u> , RESET, CS | 3 |
| 8 ✓ | Draw and explain the operational waveform of <u>8254 in MODE 0</u> operation. | 3 |
| 9 ✓ | Draw and explain the format of program status word in <u>8051</u> . | 3 |
| 10 ✓ | Write an assembly language program for 8051 to compute x to the power n where both x and n are 8-bit numbers given by user and the result should not be more than 16 bits. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- | | | |
|------|---|----|
| 11 ✓ | Draw and discuss the internal block diagram of <u>8086</u> . | 14 |
| 12 | With a neat sketch explain the read and write cycle timing diagram of 8086 in minimum mode. | 14 |

Module -2

- 13 Write an assembly language program to find the largest and smallest number from an unordered array of 16-bit numbers. Assume the array contains 15 numbers and the starting location as 2500H. Draw the flowchart for the program. 14
- 14 Write an assembly language program to find the total number of even and odd numbers from an array of 16-bit numbers. Assume the array contains 20 numbers and the starting location as 5500H. Draw the flowchart for the program. 14

Module -3

- 15 a) Explain the interrupt cycle of 8086. 8
- b) Differentiate maskable and non-maskable interrupts in 8086. 6
- 16 Draw the architectural block diagram of 8259A and explain the function of each block. 14

Module -4

- 17 Explain the different modes of operation of 8255 in detail. 14
- 18 Draw and explain the internal architecture of 8257. 14

Module -5

- 19 a) Explain the addressing modes of 8051 with example. 10
- b) Write an assembly language program for 8051 to perform addition of two 2x2 matrices. 4
- 20 a) Explain the interrupt and stack structure of 8051. 10
- b) Write an assembly language program for 8051 to find the transpose of a 2x2 matrix. 4
